

$$i) \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix}$$

Since the model has 2 predictors & 1 intercept, the design matrix has 3 columns: a 1s column for the intercept, a column for the houses, & a column for the number of bedrooms.

$$ii) \text{ Expand } S(\beta) = (y - X\beta)^T (y - X\beta) \\ \Rightarrow S(\beta) = (y^T - \beta^T X^T) (y - X\beta) \\ = y^T y - \underbrace{y^T X \beta - \beta^T X^T y}_{\text{same}} + X \beta \beta^T X^T \\ = y^T y - (y^T X \beta)^T - \beta^T X^T y + X \beta \beta^T X^T \\ = y^T y - \underbrace{\beta^T X^T y - \beta^T X^T y}_{\text{same}} + X \beta \beta^T X^T \\ = y^T y - 2 \beta^T X^T y + X \beta \beta^T X^T$$

$$S(\beta) = y^T y - 2 \beta^T X^T y - \beta^T X^T X \beta$$

iii) Differentiate $S(\beta)$ w.r.t β

$$\frac{\partial S}{\partial \beta} [y^T y - 2 \beta^T X^T y - \beta^T X^T X \beta]$$

$$\Rightarrow \frac{\partial S}{\partial \beta} [y^T y] = 0$$

$$\frac{\partial S}{\partial \beta} [-2 \beta^T X^T y] = -2 X^T y$$

$$\frac{\partial S}{\partial \beta} [-\beta^T X^T X \beta] = (X^T X + X X^T) \beta = 2 X^T X \beta$$

$$\frac{\partial [B^T A B]}{\partial B} = (A + A^T) B$$

$\therefore$

$$\frac{\partial S}{\partial \beta} = -2 X^T y + 2 X^T X \beta$$

iv) Set $\frac{\partial S}{\partial \beta} = 0$, show $\hat{\beta} = (X^T X)^{-1} X^T y$

$$0 = -2 X^T y + 2 X^T X \beta$$

$$\cancel{2} X^T y = \cancel{2} X^T X \beta$$

$$(X^T X)^{-1} X^T y = \cancel{X^T X} \beta (\cancel{X^T X})^{-1}$$

$$\hat{\beta} = (X^T X)^{-1} (X^T y)$$

v) Show $\hat{\beta}$ is unbiased estimator of β

$$\hat{\beta} = (X^T X)^{-1} (X^T y), \quad y = X \beta + \epsilon$$

$$\hat{\beta} = (X^T X)^{-1} (X^T) (X \beta + \epsilon)$$

$$= \underbrace{(X^T X)^{-1} X^T X}_{= I} \beta + (X^T X)^{-1} (X^T) (\epsilon)$$

$$\hat{\beta} = \beta + (X^T X)^{-1} X^T \epsilon$$

$$E(\hat{\beta}) = E[\beta + (X^T X)^{-1} X^T \epsilon]$$

$$= \beta + (X^T X)^{-1} X^T E(\epsilon)$$

$$= \beta + \underbrace{(X^T X)^{-1} X^T (0)}_{\rightarrow 0}, \quad E(\epsilon) = 0$$

$$\therefore E(\hat{\beta}) = \beta$$

$$vi) \quad \hat{y} = 40 + \underbrace{2.5}_{\hat{\beta}_1} x_1 + \underbrace{8}_{\hat{\beta}_2} x_2$$

For $\hat{\beta}_1$, holding everything else constant a 1 unit increase of x_1 , results in a 2.5 increase in y . i.e. A 100 square feet increase in the house size results in a \$2,500 increase in selling price.

For $\hat{\beta}_2$, holding everything else constant a 1 unit increase of x_2 , results in a 8 unit increase in y . i.e. An increase of 8 bedrooms in the house, results in an estimated increase of \$8000 in the selling price.

$$vii) \quad \hat{y} = 40 + 2.5 x_1 + 8 x_2$$

y = selling price

$\hat{\beta}_1 = 2.5$

$\hat{\beta}_2 = 8$

x_1 = Square footage in hundreds = 2000/100 = 20

x_2 = # of bedrooms = 3

$$\hat{y} = 40 + 2.5(20) + 8(3)$$

$$= 40 + 50 + 24$$

$$\hat{y} = 114$$

for a 2,000 square foot house with 3 bedrooms,

the model predicts a selling price of \$114,000